Java slip solution

Slip No 1

Q1. Write a Java program to display all the alphabets between 'A' to 'Z' after every 2 seconds

```
Public class Slip26_1 extends Thread
{
char c;
public void run()
{
for(c = 'A'; c<='Z';c++)
{
System.out.println(""+c);
try

{
Thread.sleep(3000);
}
catch(Exception e)
{
e.printStackTrace();
}
}
}
public static void main(String args[])
{
Slip26_1 t = new Slip26_1();
t.start();
}
```

2. Write a Java program to accept the details of Employee (Eno, EName, Designation, Salary) from a user and store it into the database. (Use Swing)

```
import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;
public class Ass1 extends Frame implements ActionListener
{
Label l1,l2,l3;
TextField t1,t2,t3;
Button b;
Connection cn;
Statement st;
ResultSet rs;
public Ass1()
{
setLayout(null);
l1=new Label("Eno");
l2=new Label("EName");
l3=new Label("Salary");
```

```
t1=new TextField();
t2=new TextField();
t3=new TextField();
b=new Button("Save");
l1.setBounds(50,50,100,30);
t1.setBounds(160,50,100,30);
l2.setBounds(50,90,100,30);
t2.setBounds(160,90,100,30);
l3.setBounds(50,130,100,30);
t3.setBounds(160,130,100,30);
b.setBounds(50,170,100,30);
add(l1);
add(t1);
add(l2);
add(t2);
add(l3);
add(t3);
add(b);
b.addActionListener(this);
setSize(500,500);
setVisible(true);
addWindowListener(new WindowAdapter()
{
public void windowClosing(WindowEvent e)
{
System.exit(0);
}
});
}
public void actionPerformed(ActionEvent oe)
{
String str=oe.getActionCommand();
if(str.equals("Save"))
{
try
{
Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");
cn=DriverManager.getConnection("jdbc:odbc:Ass","","");
st =cn.createStatement();
int en=Integer.parseInt(t1.getText());
String enn=t2.getText();
int sal=Integer.parseInt(t3.getText());
String strr="insert into emp values(" + en + " ,'" + enn + "'," + sal + ")";
int k=st.executeUpdate(strr);
if(k>0)
{
JOptionPane.showMessageDialog(null,"Record Is Added");
}
}
catch(Exception er)
{
```

```
System.out.println("Error");
}
}
}
public static void main(String args[])
{
new Ass1().show();
}
}
```

Slip No - 2

1.Write a java program to read 'N' names of your friends, store it into HashSet and display them in ascending order

```
// Java program to sort a HashSet

import java.util.*;

public class GFG {
	public static void main(String args[])
	{
		// Creating a HashSet
		HashSet<String> set = new HashSet<String>();

		// Adding elements into HashSet using add()
		set.add("geeks");
		set.add("practice");
		set.add("contribute");
		set.add("ide");

		System.out.println("Original HashSet: "
						+ set);

		// Sorting HashSet using List
		List<String> list = new ArrayList<String>(set);
		Collections.sort(list);

		// Print the sorted elements of the HashSet
		System.out.println("HashSet elements "
						+ "in sorted order "
						+ "using List: "
						+ list);
	}
}
```

2.Design a servlet that provides information about a HTTP request from a client, such as IP-Address and browser type. The servlet also provides information about the server on

which the servlet is running, such as the operating system type, and the names of currently loaded servlets.

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class NewServlet extends HttpServlet
{

   public void doGet(HttpServletRequest req,HttpServletResponse resp)throws
IOException,ServletException
   {
      resp.setContentType("text/html");
      String userinfo=req.getHeader("User-Agent");

      PrintWriter p=resp.getWriter();
   }
}
```

```
<html>
   <body>
      <form action="http://localhost:8080/serv/NewServlet" method="get">
     Username:<input type="text" name="t1">
                <input type="submit" >
      </form>
   </body>
</html>
```

Slip No -3

Q.1)Write a JSP program to display the details of Patient (PNo, PName, Address, age, disease) in tabular form on browser.

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<body>
<%@ page import="java.sql.*;" %>
<%! inthno;
String hname,address; %>
<%
try{
Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");

Connection cn=DriverManager.getConnection("jdbc:odbc:hospital_data","","");
Statement st=cn.createStatement();
ResultSetrs=st.executeQuery("select * from Hospital");
```

```
%>
<table border="1" width="40%"> <tr> <td>Hospital No</td> <td>Name</td>
<td>Address</td> </tr> <% while(rs.next()) { %> <tr><td><%= rs.getInt("hno") %></td>
<td><%= rs.getString("hname") %></td> <td><%= rs.getString("address") %> </tr> <%
}
cn.close();
}catch(Exception e)
{
out.println(e);
}
%>
</body>
</html>
```

Q.2)Write a Java program to create LinkedList of String objects and perform the following: i. Add element at the end of the list ii. Delete first element of the list iii. Display the contents of list in reverse order

```
import java.io.*;

// Java program to implement
// a Singly Linked List
public class LinkedList {

	Node head; // head of list

	// Linked list Node.
	// Node is a static nested class
	// so main() can access it
	static class Node {

		int data;
		Node next;

		// Constructor
		Node(int d)
		{
			data = d;
			next = null;
		}
	}

	// Method to insert a new node
	public static LinkedList insert(LinkedList list,int data)
	{
		// Create a new node with given data
		Node new_node = new Node(data);
		new_node.next = null;

		// If the Linked List is empty,
```

```
        // then make the new node as head
        if (list.head == null) {
                list.head = new_node;
        }
        else {
                // Else traverse till the last node
                // and insert the new_node there
                Node last = list.head;
                while (last.next != null) {
                        last = last.next;
                }

                // Insert the new_node at last node
                last.next = new_node;
        }

        // Return the list by head
        return list;
}

// Method to print the LinkedList.
public static void printList(LinkedList list)
{
        Node currNode = list.head;

        System.out.print("LinkedList: ");

        // Traverse through the LinkedList
        while (currNode != null) {
                // Print the data at current node
                System.out.print(currNode.data + " ");

                // Go to next node
                currNode = currNode.next;
        }

        System.out.println();
}

// **************DELETION BY KEY**************

// Method to delete a node in the LinkedList by KEY
public static LinkedList deleteByKey(LinkedList list,
                                                                int key)
{
        // Store head node
        Node currNode = list.head, prev = null;

        //
        // CASE 1:
        // If head node itself holds the key to be deleted
```

```
        if (currNode != null && currNode.data == key) {
                list.head = currNode.next; // Changed head

                // Display the message
                System.out.println(key + " found and deleted");

                // Return the updated List
                return list;
        }

        //
        // CASE 2:
        // If the key is somewhere other than at head
        //

        // Search for the key to be deleted,
        // keep track of the previous node
        // as it is needed to change currNode.next
        while (currNode != null && currNode.data != key) {
                // If currNode does not hold key
                // continue to next node
                prev = currNode;
                currNode = currNode.next;
        }

        // If the key was present, it should be at currNode
        // Therefore the currNode shall not be null
        if (currNode != null) {
                // Since the key is at currNode
                // Unlink currNode from linked list
                prev.next = currNode.next;

                // Display the message
                System.out.println(key + " found and deleted");
        }

        //
        // CASE 3: The key is not present
        //

        // If key was not present in linked list
        // currNode should be null
        if (currNode == null) {
                // Display the message
                System.out.println(key + " not found");
        }

        // return the List
        return list;
}
```

```
// **************MAIN METHOD**************

// method to create a Singly linked list with n nodes
public static void main(String[] args)
{
        /* Start with the empty list. */
        LinkedList list = new LinkedList();

        //
        // ******INSERTION******
        //

        // Insert the values
        list = insert(list, 1);
        list = insert(list, 2);
        list = insert(list, 3);
        list = insert(list, 4);
        list = insert(list, 5);
        list = insert(list, 6);
        list = insert(list, 7);
        list = insert(list, 8);

        // Print the LinkedList
        printList(list);

        //
        // ******DELETION BY KEY******
        //

        // Delete node with value 1
        // In this case the key is ***at head***
        deleteByKey(list, 1);

        // Print the LinkedList
        printList(list);

        // Delete node with value 4
        // In this case the key is present ***in the
        // middle***
        deleteByKey(list, 4);

        // Print the LinkedList
        printList(list);

        // Delete node with value 10
        // In this case the key is ***not present***
        deleteByKey(list, 10);

        // Print the LinkedList
        printList(list);
```

```
        }
}
```

Slip No - 4

Q1) Write a Java program using Runnable interface to blink Text on the frame

```
import java.awt.*;
import java.awt.event.*;

class Slip8_1 extends Frame implements Runnable
{
        Thread t;
        Label l1;
        int f;
        Slip8_1()
        {
                t=new Thread(this);
                t.start();
                setLayout(null);
                l1=new Label("Hello JAVA");
                l1.setBounds(100,100,100,40);
                add(l1);
                setSize(300,300);
                setVisible(true);
                f=0;
        }
        public void run()
        {
                try
                {
                        if(f==0)
                        {
                                t.sleep(200);
                                l1.setText("");
                                f=1;
                        }
                        if(f==1)
                        {
                                t.sleep(200);
                                l1.setText("Hello Java");
                                f=0;
                        }
                }
                catch(Exception e)
                {
                        System.out.println(e);
                }
                run();
        }
        public static void main(String a[])
```

```
        {
                new Slip8_1();
        }
}
```

Q2) Write a Java program to store city names and their STD codes using an appropriate collection and perform following operations: i. Add a new city and its code (No duplicates) ii. Remove a city from the collection iii. Search for a city name and display the code

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import java.util.*;


class Slip16_2 extends JFrame implements ActionListener
{
        JTextField t1,t2,t3;
        JButton b1,b2,b3;
        JTextArea t;
        JPanel p1,p2;

        Hashtable ts;
        Slip16_2()
        {
                ts=new Hashtable();
                t1=new JTextField(10);
                t2=new JTextField(10);
                t3=new JTextField(10);

                b1=new JButton("Add");
                b2=new JButton("Search");
                b3=new JButton("Remove");

                t=new JTextArea(20,20);
                p1=new JPanel();
                p1.add(t);

                p2= new JPanel();
                p2.setLayout(new GridLayout(2,3));
                p2.add(t1);
                p2.add(t2);
                p2.add(b1);
                p2.add(t3);
                p2.add(b2);
                p2.add(b3);

                add(p1);
```

```
            add(p2);

            b1.addActionListener(this);
            b2.addActionListener(this);
            b3.addActionListener(this);

            setLayout(new FlowLayout());
            setSize(500,500);
            setVisible(true);
            setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);


        }
        public void actionPerformed(ActionEvent e)
        {
            if(b1==e.getSource())
            {
                String name = t1.getText();
                int code = Integer.parseInt(t2.getText());
                ts.put(name,code);
                Enumeration k=ts.keys();
                Enumeration v=ts.elements();
                String msg="";
                while(k.hasMoreElements())
                {
                    msg=msg+k.nextElement()+" = "+v.nextElement()+"\n";
                }
                t.setText(msg);
                t1.setText("");
                t2.setText("");
            }
            else if(b2==e.getSource())
            {
                String name = t3.getText();

                if(ts.containsKey(name))
                {
                    t.setText(ts.get(name).toString());
                }

                else
                    JOptionPane.showMessageDialog(null,"City not found ...");
            }
            else if(b3==e.getSource())
            {
                String name = t3.getText();

                if(ts.containsKey(name))
                {
                    ts.remove(name);
                    JOptionPane.showMessageDialog(null,"City Deleted ...");
```

```
                    }

                    else
                        JOptionPane.showMessageDialog(null,"City not found ...");
                }
        }
        public static void main(String a[])
        {
                new Slip16_2();
        }
}
```

Slip No -5

Q.1)Write a Java Program to create the hash table that will maintain the mobile number and student name. Display the details of student using Enumeration interface

```
// Java Program to Demonstrate Getting Keys
// as an Enumeration of Hashtable class

// Importing required classes
import java.io.*;
import java.util.*;

// Main class
class GFG {

        // Main driver method
        public static void main(String[] args)
        {
                // Creating an empty hashtable
                Hashtable<String, String> ht
                        = new Hashtable<String, String>();

                // Inserting key-value pairs into hash table
                // using put() method
                ht.put("Name", "Rohan");
                ht.put("Mpbile_Nos", "8446049402");


                // Now creating an Enumeration object
                // to store keys
                Enumeration<String> e = ht.keys();

                // Condition holds true till there is
                // single key remaining
                while (e.hasMoreElements()) {

                        // Getting key
                        String key = e.nextElement();
```

```
                        // Printing key and value corresponding to
                        // that key
                        System.out.println(key + ":" + ht.get(key));
                }
        }
```

Q2) Create a JSP page for an online multiple choice test. The questions are randomly selected from a database and displayed on the screen. The choices are displayed using radio buttons. When the user clicks on next, the next question is displayed. When the user clicks on submit, display the total score on the screen

Exam.jsp

```
<%@page import="java.sql.*,java.util.*"%>
<%
   Class.forName("org.postgresql.Driver");

   Connection con = DriverManager.getConnection(
   "jdbc:postgresql:ty1","postgres","");

   Set s = new TreeSet();

   while(true){
      int n = (int)(Math.random()*11+1);

      s.add(n);

      if(s.size()==5) break;
   }

   PreparedStatement ps = con.prepareStatement("select * from questions where qid=?");
%>
<form method='post' action='accept_ans.jsp'>
<table width='70%' align='center'>
<%
   int i=0;
   Vector v = new Vector(s);
   session.setAttribute("qids",v);

   int qid = Integer.parseInt(v.get(i).toString());
   ps.setInt(1,qid);
   ResultSet rs = ps.executeQuery();
   rs.next();
%>
<tr>
   <td><b>Question:<%=i+1%></b></td>
</tr>
<tr>
   <td><pre><b><%=rs.getString(2)%></pre></b></td>
</tr>
<tr>
```

```
   <td>
   <b>
   <input type='radio' name='op' value=1><%=rs.getString(3)%><br>
   <input type='radio' name='op' value=2><%=rs.getString(4)%><br>
   <input type='radio' name='op' value=3><%=rs.getString(5)%><br>
   <input type='radio' name='op' value=4><%=rs.getString(6)%><br><br>
   </b>
   </td>
</tr>
<tr>
   <td align='center'>
   <input type='submit' value='Next' name='ok'>
   <input type='submit' value='Submit' name='ok'>
   </td>
</tr>
</table>
<input type='hidden' name='qno' value=<%=qid%>>
<input type='hidden' name='qid' value=<%=i+1%>>
</form>
</body>
```

Acceptans.jsp

```
<%@page import="java.sql.*,java.util.*"%>
<%
   Class.forName("org.postgresql.Driver");

   Connection con = DriverManager.getConnection(
   "jdbc:postgresql:ty1","postgres","");

   Vector answers = (Vector)session.getAttribute("answers");

   if(answers==null)
      answers = new Vector();

   int qno = Integer.parseInt(request.getParameter("qno"));
   int ans = Integer.parseInt(request.getParameter("op"));
   int i = Integer.parseInt(request.getParameter("qid"));

   answers.add(qno+" "+ans);

   session.setAttribute("answers",answers);

   String ok = request.getParameter("ok");

   if(ok.equals("Submit") || i==5){
      response.sendRedirect("result.jsp");
      return;
   }

   PreparedStatement ps = con.prepareStatement("select * from questions where qid=?");
```

```
%>
<form method='post' action='accept_ans.jsp'>
<table width='70%' align='center'>
<%
   Vector v = (Vector)session.getAttribute("qids");

   int qid = Integer.parseInt(v.get(i).toString());
   ps.setInt(1,qid);
   ResultSet rs = ps.executeQuery();
   rs.next();
%>
<tr>
<td><b>Question:<%=i+1%></b></td>
</tr>
<tr>
<td><pre><b><%=rs.getString(2)%></pre></b></td>
</tr>
<tr>
<td>
<b>
<input type='radio' name='op' value=1><%=rs.getString(3)%><br>
<input type='radio' name='op' value=2><%=rs.getString(4)%><br>
<input type='radio' name='op' value=3><%=rs.getString(5)%><br>
<input type='radio' name='op' value=4><%=rs.getString(6)%><br><br>
</b>
</td>
</tr>
<tr>
   <td align='center'>
   <input type='submit' value='Next' name='ok'>
   <input type='submit' value='Submit' name='ok'>
   </td>
</tr>
</table>
<input type='hidden' name='qno' value=<%=qid%>>
<input type='hidden' name='qid' value=<%=i+1%>>
</form>
</body>
```

Result.jsp

```
<%@page import="java.sql.*,java.util.*,java.text.*"%>
<%
   Class.forName("org.postgresql.Driver");

   Connection con = DriverManager.getConnection(
   "jdbc:postgresql:ty1","postgres","");

   Vector v = (Vector)session.getAttribute("answers");
   if(v==null){
%>
```

```
<h1>No questions answered</h1>
<%
      return;
   }

   PreparedStatement ps = con.prepareStatement("select ans from questions where
qid=?");

   int tot=0;

   for(int i=0;i<v.size();i++){
      String str = v.get(i).toString();
      int j = str.indexOf(' ');
      int qno = Integer.parseInt(str.substring(0,j));
      int gans = Integer.parseInt(str.substring(j+1));

      ps.setInt(1,qno);

      ResultSet rs = ps.executeQuery();
      rs.next();

      int cans = rs.getInt(1);

      if(gans==cans) tot++;
   }

   session.removeAttribute("qids");
   session.removeAttribute("answers");
   session.removeAttribute("qid");
%>
<h3>Score:<%=tot%></h1>
<center><a href='exam.jsp'>Restart</a></center>
</body>
```

SQL File

```
create table questions(qid serial primary key, question text, option1 text, option2 text,
option3 text, option4 text, ans int);

insert into questions
(question,option1,option2,option3,option4,ans)
values
('Who is prime minister of India?','Rahul Gandhi','Narendra Modi','Sonia
Gandhi','Manmohan Singh',2),
('Who is finance minister of India','Rahul Gandhi','P Chidambaram','Manmohan
Singh','Arun Jately',4),
('What is square root of 16?','2','4','1','256',4),
('Who is chief minister of Maharashtra','Uddhav Tharakey','Devendra Fadanavis','Raj
Thakarey','Sharad Pawar',2),
('What is full for of LIFO?','Last In First Out','Late In First Out','Long In First Out','Large In
First Out',1),
('Which is capital of India','Delhi','Maharashtra','Kolkata','Goa',1), ('What is currency of
India','Dollar','Rupee','Pound','Yen',2),
```

('Who Invented C?','Kim Thompson','Bill Joy','Dennis Ritche','Balaguru Swamy',3),
('Where was Java invented?','Microsoft','Oracle','Sun Microsystem','Intel',3),
('What is cube root of 8?','2','3','4','5',1),('What is full form of FIFO','Fast In Fast Out','First in First Out','Fast In First Out','First In Fast Out',2);

Slip No -6

Q1) Write a Java program to accept 'n' integers from the user and store them in a collection. Display them in the sorted order. The collection should not accept duplicate elements. (Use a suitable collection). Search for a particular element using predefined search method in the Collection framework.

```
import java.util.*;
import java.io.*;

class Slip19_2
{
	public static void main(String[] args) throws Exception
	{
		int no,element,i;
			BufferedReader br=new BufferedReader(new
InputStreamReader(System.in));
			TreeSet ts=new TreeSet();
			System.out.println("Enter the of elements :");
			no=Integer.parseInt(br.readLine());
			for(i=0;i<no;i++)
			{
				System.out.println("Enter the element : ");
					element=Integer.parseInt(br.readLine());
					ts.add(element);
			}

			System.out.println("The elements in sorted order :"+ts);
		System.out.println("Enter element to be serach : ");
		element = Integer.parseInt(br.readLine());
		if(ts.contains(element))
			System.out.println("Element is found");
		else
			System.out.println("Element is NOT found");
	}
}
```

Q2) Write a java program to simulate traffic signal using threads

```
import java.applet.*;
import java.awt.*;
```

```
class Slip3_2 extends Applet implements Runnable
{
Thread t;
int r,g1,y,i;
public void init()
{
T=new Thread(this);
t.start();
r=0; g1=0;I=0; y=0;
}
public void run()
{
try
{
for(I =24; I >=1;i--)
{
if (I >16&& I <=24)
{
t.sleep(200);
r=1;
repaint();
}
if (I >8&& I <=16)
{
t.sleep(200);
y=1;
repaint();
}
if(I >1&& I <=8)
{
t.sleep(200);
g1=1;
repaint();
}
}
if (I ==0)
{
run();
}
}
catch(Exception e)
{ System.out.println(e);
}
} public void paint(Graphics g)
{
g.drawRect(100,100,100,300);
if (r==1)
{
g.setColor(Color.red);
g.fillOval(100,100,100,100);
```

```
g.setColor(Color.black);
g.drawOval(100,200,100,100);
g.drawOval(100,300,100,100);
r=0;
}
if (y==1)
{
g.setColor(Color.black);
g.drawOval(100,100,100,100);
g.drawOval(100,300,100,100);
g.setColor(Color.yellow);
g.fillOval(100,200,100,100);
y=0;
}
if (g1==1)
{
g.setColor(Color.black);
g.drawOval(100,100,100,100);
g.drawOval(100,200,100,100);
g.setColor(Color.green);
g.fillOval(100,300,100,100);
g1=0;
}
}
}
```

Slip No -7

Q1. Write a java program that implements a multi-thread application that has three threads. First thread generates random integer number after every one second, if the number is even; second thread computes the square of that number and print it. If the number is odd, the third thread computes the of cube of that number and print it.

```
import java.util.Random;
class Square extends Thread
{
 int x;
 Square(int n)
 {
 x = n;
 }
 public void run()
 {
 int sqr = x * x;
 System.out.println("Square of " + x + " = " + sqr );
 }
}
class Cube extends Thread
```

```
{
 int x;
 Cube(int n)
 {x = n;
 }
 public void run()
 {
 int cub = x * x * x;
 System.out.println("Cube of " + x + " = " + cub );
 }
}
class Number extends Thread
{
 public void run()
 {
 Random random = new Random();
 for(int i =0; i<5; i++)
 {
 int randomInteger = random.nextInt(100);
 System.out.println("Random Integer generated : " + randomInteger);
 Square s = new Square(randomInteger);
 s.start();
 Cube c = new Cube(randomInteger);
 c.start();
 try {
 Thread.sleep(1000);
} catch (InterruptedException ex) {
 System.out.println(ex);
}
 }
 }
}
public class Thr {
 public static void main(String args[])
 {
 Number n = new Number();
 n.start();
 }
}
```

Q2. Write a java program for the following: i. To create a Product(Pid, Pname, Price) table. ii. Insert at least five records into the table. iii. Display all the records from a table.

```
// Java program to illustrate
// inserting to the Database
import java.sql.*;

public class insert1
{
	public static void main(String args[])
	{
		String id = "id1";
		String pwd = "pwd1";
		String fullname = "geeks for geeks";
		String email = "geeks@geeks.org";

		try
		{
			Class.forName("oracle.jdbc.driver.OracleDriver");
			Connection con = DriverManager.getConnection("
			jdbc:oracle:thin:@localhost:1521:orcl", "login1", "pwd1");
			Statement stmt = con.createStatement();

			// Inserting data in database
			String q1 = "insert into userid values('" +id+ "', '" +pwd+
								"', '" +fullname+ "', '"
+email+ "')";
			int x = stmt.executeUpdate(q1);
			if (x > 0)
				System.out.println("Successfully Inserted");
			else
				System.out.println("Insert Failed");

			con.close();
		}
		catch(Exception e)
		{
			System.out.println(e);
		}
	}
}
```

Slip No -8

Q.1)Write a java program to define a thread for printing text on output screen for 'n' number of times. Create 3 threads and run them. Pass the text 'n' parameters to the thread constructor. Example: i. First thread prints "COVID19" 10 times. ii. Second thread prints "LOCKDOWN2020" 20 times iii. Third thread prints "VACCINATED2021" 30 times

```
public class A1 extends Thread {
   String str;
   int n;
```

```
    A1(String str, int n) {
        this.str = str;
        this.n = n;
    }

    public void run() {
        try {
            for (int i = 0; i < n; i++) {
                System.out.println(getName() + " : " + str);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
    public static void main(String[] args) {
        A1 t1 = new A1("COVID19", 10);
        A1 t2 = new A1("LOCKDOWN2020", 20);
        A1 t3 = new A1("VACCINATED", 30);

        t1.start();
        t2.start();
        t3.start();

    }
}
```

2.Write a JSP program to check whether a given number is prime or not. Display the result in red color.

```
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body><center><h1>The required Result is:: </h1>
        <h2>
            <%
            int n,i,flag=0;

            String ns= request.getParameter("n");
            n=Integer.parseInt(ns);
            if(n>1)
                {

                for(i=2;i<=n/2;i++)
                    {
                        if(n%i==0)
                            {
                                flag=1;
                                break;
```

```
            }
          }
        }
      if(flag==0)
        {
        out.println("<pre>");
      out.println(n+" is a prime no.");
       out.println("</pre>");
      }
      else
        {
         out.println("<pre>");
         out.println(n+" is not a prime no.");
          out.println("</pre>");
      }

      %>

      </h2></center>
  </body>
</html>
```

Slip No - 9

1. Write a Java program to create a thread for moving a ball inside a panel vertically. The ball should be created when the user clicks on the start button

```
import java.awt.*;import java.awt.geom.*;import javax.swing.*;import
java.awt.event.*;import java.util.*;
public class BouncingBallApp extends JFrame
{
   //start of main method
   public static void main(String[] args)
   {
     //crate container
     Container container = new Container();
     //crate BouncingBallApp instance
     BouncingBallApp bBalls = new BouncingBallApp();
      //set the window closing feature(close with X click)
     bBalls.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
     //crate boucing ball panel(BBP) instance and add
     BouncingBallPanel BBP = new BouncingBallPanel();
     container.add(BBP);
     //make the BBP the MouseListner
     bBalls.addMouseListener(BBP);
     //set window background and size
     bBalls.setBackground(Color.WHITE);
     bBalls.setSize(400, 300);
```

```
      BBP.setSize(400, 300);
      BBP.setLayout(null);
      bBalls.setContentPane(BBP);
      //set window visible
      bBalls.setVisible(true);
    }//end of main method
}//end of Class BouncingBall App

class BouncingBallPanel extends JPanel implements MouseListener
{
    //create an empty array for 20 Ball objects
    public Ball[] array;
    private int count = 0;
    Random generator = new Random();

    public BouncingBallPanel()
    {
        array = new Ball[20];
    }

        public void mouseClicked(MouseEvent event)
    {
        array[count] = new Ball(this);
        count++;
        if( count == 1)
        {
            final Runnable update = new Runnable()
            {
                public void run()
                {
                    for (int j = 0; j < array.length; j++)
                    {
                        if(array[j] != null)
                        {
                            array[j].move();
                        }//end of if
                    }//end of for
                }//end of run method
            };//end of runnalbe update
            (new Thread(new Ball(this))).start();
            Runnable graphic = new Runnable()
            {
                public void run()
                {
                    while(true)
                    {
                        try
                        {
                            EventQueue.invokeLater(update);
                            Thread.sleep(generator.nextInt(10 +100));
                        }catch (InterruptedException exp){}
```

```
            }//end of while
          }//end of run
        };//end of runnable
        new Thread(graphic).start();
      }//end of if
    }//end of mouseClicked method

    //empty interfaces for mouse events
    public void mouseExited(MouseEvent event){}
    public void mouseReleased(MouseEvent event){}
    public void mouseEntered(MouseEvent event){}
    public void mousePressed(MouseEvent event){}

    //paint component method
    public void paintComponent(Graphics g)
    {
     super.paintComponent(g);
     Graphics2D g2d = (Graphics2D) g;
     //loop for each ball and draw all balls in array
     for(int i = 0; i < array.length; i++)
     {
       if(array[i] != null)
       {
        g2d.setColor(array[i].getColor());
        g2d.fillOval((int)array[i].getX(), (int)array[i].getY(), (int)array[i].getDiameter(),
(int)array[i].getDiameter());
       }
     }//end of for loop
    }//end of paintComponent loop
}//end of Class BouncingBallPanel
class Ball implements Runnable
{
    //set up variables
    private double x;
    private double y;
    private int deltaX;
    private int deltaY;
    private double diameter;
    private Color color;
    BouncingBallPanel BBP2;

    Random random = new Random();

    public Ball(BouncingBallPanel a)
    {
       x = random.nextInt(400);
       y = random.nextInt(300);
       deltaX = 1 + random.nextInt(10);
       deltaY = 1 + random.nextInt(10);
       diameter = 5 + random.nextInt(20);
       color = new Color(random.nextInt(256), random.nextInt(256), random.nextInt(256));
```

```
        BBP2 = a;
    }// end of constructor

    public double getX()
    {
        return x;
    }

    public double getY() {
            return y;
    }

    public double getDiameter() {
        return diameter;
    }

    public Color getColor() {
        return color;
    }

    public void move() {
        x += deltaX;
        y += deltaY;

        if (x > 400 - getDiameter()|| x <0)
        {
            deltaX = -deltaX;
        }

        if (y > 300 - getDiameter() || y < 0)
        {
            deltaY = -deltaY;
        }

    }// end of method move
    @Override
    public void run()
    {
        while(true)
         {
            move();
            BBP2.repaint();
            try{
              Thread.currentThread().sleep(10 + random.nextInt(100));
            }catch(InterruptedException exp){}
           }//end of while
    }//end of run method
}//end of Ball
```

2.Write a Java program using Spring to display the message “If you can't explain it simply, you don't understand it well enough”
Ans:-

```
// Step 1: Create a new Spring project in your IDE.

// Step 2: Add the required dependencies to your pom.xml file.
// Here's an example:
/*
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-webmvc</artifactId>
    <version>5.3.8</version>
  </dependency>
</dependencies>
*/

// Step 3: Create a new class called MainController.
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.ResponseBody;

@Controller
public class MainController {

  // Step 6: Create a method that returns the message as a string.
  private String message = "If you can't explain it simply, you don't understand it well
enough.";

  // Step 5: Use the @RequestMapping annotation to map a URL path to this controller.
  @RequestMapping("/message")
  // Step 7: Use the @ResponseBody annotation to indicate that the method's return
value should be sent as the HTTP response body.
  @ResponseBody
  public String showMessage() {
    return message;
  }
}

// Step 8: Run the Spring application and navigate to the URL mapped in step 5 to see the
message.
// For example, if you're running the application on localhost:8080, navigate to
http://localhost:8080/message to see the message.
```

Slip No -10

Q1) Write a Java program to display the Current Date using spring

```
// Java Program to Display Current Date and Time
import java.text.*;
import java.util.*;
public class GFG {
	public static void main(String args[])
	{
		SimpleDateFormat formatDate = new SimpleDateFormat(
			"dd/MM/yyyy HH:mm:ss z");
		//"SimpleDateFormat" class initialize with object
		//"formatDate" this class acceptes the format of
		// date and time as ""dd/MM/yyyy" and "HH:mm:ss z""
		//"z" use for print the time zone

		Date date = new Date();
		// initialize "Date" class

		formatDate.setTimeZone(TimeZone.getTimeZone("IST"));
		// converting to IST or format the Date as IST

		System.out.println(formatDate.format(date));
		// print formatted date and time
	}
}
```

Q2)
Write a Java program to display first record from student table (RNo, SName, Per) onto the TextFields by clicking on button. (Assume Student table is already created).

```
$ sudo -u postgres psql
[sudo] password for codeforever:
psql (9.3.11)
Type “help” for help.
postgres=# create database stud;
CREATE DATABASE
postgres=# \c stud
You are now connected to database “stud” as user “postgres”.
stud=# create table student(rollno int primary key,name text,percentage float);
CREATE TABLE
*/


package studdb;
```

```
import javax.swing.table.*;
import java.sql.*;
import java.awt.event.*;
import java.awt.*;
import javax.swing.*;
public class StudDb extends JFrame implements ActionListener
{
   JTextField t1,t2,t3;
   JLabel l1,l2,l3;
   JButton b1,b2;
   int row,column;
   StudDb()
   {
      setLayout(new FlowLayout());
      setSize(500,500);
      setVisible(true);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      l1=new JLabel("RollNo");
      add(l1);

      t1=new JTextField(10);
      add(t1);

      l2=new JLabel("Name");
      add(l2);

      t2=new JTextField(10);
      add(t2);

      l3=new JLabel("Percentage");
      add(l3);

      t3=new JTextField(10);
      add(t3);

      b1=new JButton("Insert");
      add(b1);
      b1.addActionListener(this);

      b2=new JButton("Display");
      add(b2);
      b2.addActionListener(this);

      try
      {
         Class.forName("org.postgresql.Driver");
      }
      catch(Exception e)
      {
        System.out.println("Error"+e.getMessage());
```

```
      }
   }
   public void actionPerformed(ActionEvent e2)
   {
      if(e2.getSource()==b1)
      {
         try
        {
            int eno=Integer.parseInt(t1.getText());
            String ename=t2.getText();
            float percentage=Float.parseFloat(t3.getText());

    Connection conn =
DriverManager.getConnection("jdbc:postgresql://localhost/stud","postgres","password");

             PreparedStatement st=conn.prepareStatement("insert into student
values(?,?,?)");
             st.setInt(1, eno);
             st.setString(2,ename);
             st.setFloat(3,percentage);
             st.executeUpdate();
             st.close();
             JOptionPane.showMessageDialog(this,"Inserted");

        }catch(Exception e)
        {
            System.out.println("Error"+e.getMessage());
        }
      }
      if(e2.getSource()==b2)
      {
         try
        {

            Object[] data=new Object[3];
            DefaultTableModel dtm=new DefaultTableModel();
            JTable jt=new JTable(dtm);
            ResultSet rs;
             System.out.println("hello");
            jt.setBounds(20,20,50,50);
            String[] darr={"RollNo","Name","Percentage"};
            for(int column=0;column<3;column++)
            {
               dtm.addColumn(darr[column]);
            }
            Connection conn =
DriverManager.getConnection("jdbc:postgresql://localhost/stud","postgres","password");
             Statement st=conn.createStatement();
             rs=st.executeQuery("select * from student");
            for(row=0;rs.next();row++)
            {
```

```
            for(int column=0;column<3;column++)
            {
               data[column]=rs.getObject(column+1);
            }
            dtm.addRow(data);
          }
          rs.close();
          getContentPane().add(new JScrollPane(jt));
        }catch(Exception e)
        {

        }
      }
   }
   public static void main(String[] args)
   {
      new StudDb();
   }

}
```

Slip No -11

Design an HTML page which passes customer number to a search servlet. The servlet searches for the customer number in a database (customer table) and returns customer details if found the number otherwise display error message.

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.sql.*;
public class servletDatabase extends HttpServlet
{
   Connection cn;
   public void init()
   {
     try
     {
         Class.forName("org.gjt.mm.mysql.Driver");
         cn=DriverManager.getConnection("jdbc:mysql://localhost/stud","root","password"
);
         System.out.println("Hii");
     }
     catch(Exception ce)
     {
        System.out.println("Error"+ce.getMessage());
     }

   }
   public void doGet(HttpServletRequest req, HttpServletResponse resp)
         throws ServletException, IOException
```

```
  {
    resp.setContentType("text/html");
    PrintWriter pw=resp.getWriter();
    try
    {
      int rno=Integer.parseInt(req.getParameter("t1"));
      String qry="Select * from student where rollno="+rno;
      Statement st=cn.createStatement();
      ResultSet rs=st.executeQuery(qry);
      while(rs.next())
      {
        pw.print("<table border=1>");
        pw.print("<tr>");
        pw.print("<td>" + rs.getInt(1) + "</td>");
        pw.print("<td>" + rs.getString(2) + "</td>");
        pw.print("<td>" + rs.getFloat(3) + "</td>");
        pw.print("</tr>");
        pw.print("</table>");
      }
    }
    catch(Exception se){}
    pw.close();
  }
}
```

HTML File

```
<html>
  <body>
    <form action="http://localhost:8080/servDb/servletDatabase" method="get">
      Enter Roll No:<input type="text" name="t1">
      <input type="submit">
    </form>
  </body>
</html>
```

```
pssql> create database stud;
Query OK, 1 row affected (0.00 sec)

pssql> create table student(rollno int primary key,name text,percentage float);
Query OK, 0 rows affected (0.07 sec)
pssql> insert into student values(1,'student1',79);
Query OK, 1 row affected (0.04 sec)
pssql> insert into student values(2,'student2',69);
Query OK, 1 row affected (0.05 sec)
pssql> insert into student values(3,'student3',58);
Query OK, 1 row affected (0.06 sec)

pssql> select * from student;
```

2.Write a Java program to display information about all columns in the DONAR table using ResultSetMetaData.

```
import java.sql.*;
import java.io.*;
public class ResultSetMetaData
{
  public static void main(String[] args) throws Exception
  {


    Statement stmt;
     Class.forName("org.postgresql.Driver");
       Connection conn =
DriverManager.getConnection("jdbc:postgresql://localhost/stud","postgres","password");
    stmt = conn.createStatement();
   ResultSet rs = stmt.executeQuery("Select * from student");
    java.sql.ResultSetMetaData rsmd = rs.getMetaData();
    int noOfColumns = rsmd.getColumnCount();
    System.out.println("Number of columns = " + noOfColumns);
    for(int i=1; i<=noOfColumns; i++)
    {
       System.out.println("Column No : " + i);
       System.out.println("Column Name : " + rsmd.getColumnName(i));
       System.out.println("Column Type : " + rsmd.getColumnTypeName(i));
      System.out.println("Column display size : " + rsmd.getColumnDisplaySize(i));
    }
    conn.close();
  }
}
```

Slip No -12

Q.1)Write a JSP program to check whether given number is Perfect or not. (Use Include directive)

Index.html file:

```
<!DOCTYPE html>
<html>
<head>
<title>PERFECT NUMBER</title>
</head>
<body>
<form action="perfect.jsp" method="post">
Enter Number :<input type="text" name="num">
<input type="submit" value="Submit" name="s1">
</form>
</body>
</html>
```

Perfect.jsp file:

```
<%@ page import="java.util.*" %>

<%
if(request.getParameter("s1")!=null)
  {
Integer num,a,i,sum = 0;
num = Integer.parseInt(request.getParameter("num"));
a = num;

for(i=1;i<a;i++)
{
if(a%i==0)
{
sum=sum + i;
}
}
if(sum==a)
{
out.println(+num+ "is a perfect number");
}
else
{
out.println(+num+ "is not a perfect number");
}
  }
%>
```

Q2) Write a Java Program to create a PROJECT table with field's project_id, Project_name, Project_description, Project_Status. Insert values in the table. Display all the details of the PROJECT table in a tabular format on the screen.(using swing)

```
import java.sql.*;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import java.util.*;


class Slip13_2 extends JFrame implements ActionListener
{
        JLabel l1,l2,l3;
        JTextField t1,t2,t3;
        JButton b1,b2,b3;
        String sql;
        JPanel p,p1;
        Connection con;
```

```
        PreparedStatement ps;


        JTable t;
        JScrollPane js;
        Statement stmt ;
        ResultSet rs ;
        ResultSetMetaData rsmd ;
        int columns;
        Vector columnNames = new Vector();
        Vector data = new Vector();

        Slip13_2()
        {

                l1 = new JLabel("Enter no :");
                l2 = new JLabel("Enter name :");
                l3 = new JLabel("percentage :");

                t1 = new JTextField(20);
                t2 = new JTextField(20);
                t3 = new JTextField(20);

                b1 = new JButton("Save");
                b2 = new JButton("Display");
                b3 = new JButton("Clear");

                b1.addActionListener(this);
                b2.addActionListener(this);
                b3.addActionListener(this);

                p=new JPanel();
                p1=new JPanel();
                p.add(l1);
                p.add(t1);
                p.add(l2);
                p.add(t2);
                p.add(l3);
                p.add(t3);

                p.add(b1);
                p.add(b2);
                p.add(b3);

                add(p);
                setLayout(new GridLayout(2,1));
                setSize(600,800);
                setVisible(true);
                setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

```
        }

        public void actionPerformed(ActionEvent e)
        {
                if((JButton)b1==e.getSource())
                {
                        int no = Integer.parseInt(t1.getText());
                        String name = t2.getText();
                        int p = Integer.parseInt(t3.getText());
                        System.out.println("Accept Values");
                        try
                        {
                                Class.forName("org.postgresql.Driver");
con=DriverManager.getConnection("jdbc:postgresql://192.168.100.254/Bill","oracle","ora
cle");
sql = "insert into stud values(?,?,?)";
                                ps = con.prepareStatement(sql);
                                ps.setInt(1,no);
                                ps.setString(2, name);
                                ps.setInt(3,p);
                                System.out.println("values set");
                                int n=ps.executeUpdate();
                                if(n!=0)
                                {
                                        JOptionPane.showMessageDialog(null,"Record insered
...");
                                }

                                else
                                        JOptionPane.showMessageDialog(null,"Record NOT
inserted ");

                        }//end of try
                        catch(Exception ex)
                        {
                                System.out.println(ex);
                                //ex.printStackTrace();
                        }

                }//end of if
                else if((JButton)b2==e.getSource())
                {
                        try
                        {
                                Class.forName("org.postgresql.Driver");
con=DriverManager.getConnection("jdbc:postgresql://192.168.100.254/Bill","oracle","ora
cle");
                                System.out.println("Connected");
                                stmt=con.createStatement();
                                rs = stmt.executeQuery("select * from stud");
                                rsmd = rs.getMetaData();
```

```
                        columns = rsmd.getColumnCount();

                        //Get Columns name
                        for(int i = 1; i <= columns; i++)
                        {
                                columnNames.addElement(rsmd.getColumnName(i));
                        }

                        //Get row data
                        while(rs.next())
                        {
                                Vector row = new Vector(columns);
                                for(int i = 1; i <= columns; i++)
                                {
                                        row.addElement(rs.getObject(i));
                                }
                                data.addElement(row);
                        }

                        t = new JTable(data, columnNames);
                        js = new JScrollPane(t);

                        p1.add(js);
                        add(p1);

                        setSize(600, 600);
                        setVisible(true);
                }
                catch(Exception e1)
                {
                        System.out.println(e1);
                }
        }
        else
        {
                t1.setText(" ");
                t2.setText(" ");
                t3.setText(" ");

        }
    }//end of method

    public static void main(String a[])
    {
        Slip13_2 ob = new Slip13_2();
    }
}
```

Slip No 13

Q1) Write a Java program to display information about the database and list all the tables in the database. (Use DatabaseMetaData).

```
import java.sql.*;
import java.io.*;
public class DBMetaData
{
  public static void main(String[] args) throws Exception
  {
    ResultSet rs = null;
     Class.forName("org.postgresql.Driver");
      Connection conn =
DriverManager.getConnection("jdbc:postgresql://localhost/dbtry","postgres","redhat");
   DatabaseMetaData dbmd = conn.getMetaData();
   System.out.println("Database Product name = " + dbmd.getDatabaseProductName());
   System.out.println("User name = " + dbmd.getUserName());
   System.out.println("Database driver  name= " + dbmd.getDriverName());
   System.out.println("Database driver version = "+ dbmd.getDriverVersion());
   System.out.println("Database product name = " + dbmd.getDatabaseProductName());
   System.out.println("Database Version = " + dbmd.getDriverMajorVersion());
   rs = dbmd.getTables(null,null,null, new String[]{"TABLE"});
   System.out.println("List of tables...");
   while(rs.next())
   {
       String tblName = rs.getString("TABLE_NAME");
       System.out.println("Table : "+ tblName);
   }
   conn.close();
  }
}
```

Q2) Write a Java program to show lifecycle (creation, sleep, and dead) of a thread. Program should print randomly the name of thread and value of sleep time. The name of the thread should be hard coded through constructor. The sleep time of a thread will be a random integer in the range 0 to 4999.

```
Class MyThread extends Thread
{ public MyThread(String s)
{
super(s);
}
public void run()
{
System.out.println(getName()+"thread created.");
while(true)
{
System.out.println(this);
int s=(int)(math.random()*5000);
System.out.println(getName()+"is sleeping for :+s+"msec");
```

```
try{
Thread.sleep(s);
}
catch(Exception e)
{
}
}
}
Class ThreadLifeCycle
{
public static void main(String args[])
{
MyThread t1=new MyThread("shradha"),t2=new MyThread("pooja");
t1.start();
t2.start();
try
{
t1.join();
t2.join();
}
catch(Exception e)
{
}
System.out.println(t1.getName()+"thread dead.");
System.out.println(t2.getName()+"thread dead.");
}
}
```

Slip No 14

Q1) Write a Java program for a simple search engine. Accept a string to be searched. Search the string in all text files in the current folder. Use a separate thread for each file. The result should display the filename and line number where the string is found.

```
import java.io.*;
public class SearchThread extends Thread
{
   File f1;
   String fname;
   static String str;
   String line;
    LineNumberReader reader = null;
   SearchThread(String fname)
   {
      this.fname=fname;
      f1=new File(fname);
   }
   public void run()
   {
```

```
      try
      {
         FileReader fr=new FileReader(f1);
         reader=new  LineNumberReader(fr);
         while((line=reader.readLine())!=null)
         {
            if(line.indexOf(str)!=-1)
            {
               System.out.println("string found in "+fname+"at
"+reader.getLineNumber()+"line");
               stop();
            }
         }
      }
      catch(Exception e)
      {
      }
   }
   public static void main(String[] args) throws IOException
   {
      Thread t[]=new Thread[20];
      BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
      System.out.println("Enter String to search");
      str=br.readLine();

      FilenameFilter filter = new FilenameFilter()
      {
         public boolean accept(File file, String name)
      {
            if (name.endsWith(".txt"))
         {
               return true;
            }
         else
            {
             return false;
            }
         }
   };

   File dir1 = new File(".");
   File[] files = dir1.listFiles(filter);
      if (files.length == 0)
   {
      System.out.println("no files available with this extension");
   }
   else
      {
          for(int i=0;i<files.length;i++)
          {
             for (File aFile : files)
```

```
                {
                  t[i]=new SearchThread(aFile.getName());
                  t[i].start();
                   }
            }
        }
    }
}
```

Q2) Write a JSP program to calculate sum of first and last digit of a given number. Display sum in Red Color with font size 18.

HTML FILE

```
<!DOCTYPE html>
<html>
<body>
<form method=post action="Slip7.jsp">
Enter Any Number : <Input type=text name=num><br><br>
<input type=submit value=Display>
</form>
</body>
</html>
```

JSP FILE:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<body>
<%! int n,rem,r; %>
<% n=Integer.parseInt(request.getParameter("num"));
     if(n<10)
    {
      out.println("Sum of first and last digit is   ");
%><font size=18 color=red><%= n %></font>
<%
    }
   else
   {
     rem=n%10;
     do{
             r=n%10;
             n=n/10;
          }while(n>0);
       n=rem+r;
      out.println("Sum of first and last digit is    ");
%><font size=18 color=red><%= n %></font>
<%
```

```
   }
%>
</body>
</html>
```

Slip No 15

Q1) Write a java program to display name and priority of a Thread

```
public class MainThread
{
   public static void main(String arg[])
   {
      Thread t=Thread.currentThread();
      System.out.println("Current Thread:"+t);//Change Name       t.setName("My Thread
");
      System.out.println ("After the name is Changed:"+t);
      try      {
         for(int i=2;i>0;i--)
         {
            System.out.println(i);
            Thread.sleep(1000);
         }
      }
      catch(Exception e)
      {
         System.out.println(e);
      }
   }
}
```

Q2) Write a SERVLET program which counts how many times a user has visited a web page. If user is visiting the page for the first time, display a welcome message. If the user is revisiting the page, display the number of times visited. (Use Cookie)

```
 import java.io.*;
      import javax.servlet.*;
      import javax.servlet.http.*;public class VisitServlet extends HttpServlet
{
   static int i=1;
   public void doGet(HttpServletRequest request,HttpServletResponse response)
         throws IOException,ServletException
   {
      response.setContentType("text/html");
      PrintWriter out=response.getWriter();
      String k=String.valueOf(i);
      Cookie c=new Cookie("visit",k);
      response.addCookie(c);
      int j=Integer.parseInt(c.getValue());
      if(j==1)
      {
```

```
        out.println("Welcome to web page ");
      }
      else      {
        out.println("You are visited at "+i+" times");
      }
      i++;
   }
}Web.xml
<?xml version="1.0" encoding="ISO-8859-1"?>
<web-app>
<servlet>
<servlet-name>VisitServlet</servlet-name>
<servlet-class>VisitServlet</servlet-class>
</servlet>
      36<servlet-mapping>
<servlet-name>VisitServlet</servlet-name>
<url-pattern>/VS</url-pattern>
</servlet-mapping>
</web-app>
```

Slip No -16

Q.1)Write a java program to create a TreeSet, add some colors (String) and print out the content of TreeSet in ascending order.

```
import java.util.*;

public class S16Q1 {

public static void main(String args[]) {

Scanner sc = new Scanner(System.in);

TreeSet<String> colors = new TreeSet<>();

System.out.print("\nEnter the number of colors: ");

int n = sc.nextInt();

sc.nextLine();

for(int i=0; i<n; i++) {

System.out.print("Enter color " + (i+1) + ": ");

colors.add(sc.nextLine());

}
```

```
System.out.println("\nTreeSet in ascending order is: " + colors);

sc.close();

}

}
```

Q.2) Write a Java program to accept the details of Teacher (TNo, TName, Subject). Insert at least 5 Records into Teacher Table and display the details of Teacher who is teaching â€œJAVAâ€ Subject. (Use PreparedStatement Interface)

```
import java.sql.*;

public class S16Q2 {

public static void main(String args[]) {

try {

Class.forName("org.postgresql.Driver");

Connection con =

DriverManager.getConnection("jdbc:postgresql://localhost:5432/ty92","ty92","ty

92");
Statement stmt = con.createStatement();

PreparedStatement pstmt = null;

stmt.executeUpdate("CREATE TABLE IF NOT EXISTS Teacher (TNO int

primary key, TNAME varchar(20), SUBJECT varchar(20))");

System.out.println("Table exists / created!");

System.out.println("Inserting values into the table...");

pstmt = con.prepareStatement("INSERT INTO

Teacher(TNO,TNAME,SUBJECT) VALUES (?,?,?)");

insertTeacher(pstmt, 101, "Taskar", "JAVA");

insertTeacher(pstmt, 102, "Mahale", "DSA");
```

```
insertTeacher(pstmt, 103, "Deore", "C");

insertTeacher(pstmt, 104, "Patil", "CN");

insertTeacher(pstmt, 105, "Kapse", "OS");

System.out.println("\nRecords inserted successfully!");

System.out.println("\nDisplaying details of teachers teaching

'JAVA' subject...");

pstmt = con.prepareStatement("SELECT * FROM Teacher WHERE SUBJECT

= ?");

pstmt.setString(1,"JAVA");

ResultSet rs = pstmt.executeQuery();

while(rs.next()) {

int tno = rs.getInt("TNO");

String tname = rs.getString("TNAME");

String sub = rs.getString("SUBJECT");

System.out.println("\nTeacher No.: " + tno);

System.out.println("Teacher Name: " + tname);

System.out.println("Subject: " + sub);

}

rs.close();

pstmt.close();

con.close();

} catch (SQLException | ClassNotFoundException e) {

System.out.println(e);

}

}
private static void insertTeacher(PreparedStatement pstmt, int tno, String
```

```
tname, String sub) throws SQLException {

pstmt.setInt(1, tno);

pstmt.setString(2, tname);

pstmt.setString(3, sub);

pstmt.executeUpdate();

}

}
```

Slip No- 17

Q.1) Write a java program to accept â€˜Nâ€™ integers from a user. Store and display integers in sorted order having proper collection class. The collection should not accept duplicate elements.

```
import java.util.*;

public class S17Q1 {

public static void main(String args[]) {

Scanner sc = new Scanner(System.in);

TreeSet<Integer> set = new TreeSet<>();

System.out.print("\nEnter the number of integers: ");

int n = sc.nextInt();

for(int i=0; i<n; i++) {

System.out.print("Enter integer " + (i+1) + ": ");

set.add(sc.nextInt());

}

System.out.println("\nSorted set without duplicates:-");

for(int num : set)

System.out.println(num);
```

```
sc.close();

}

}
```

Q.2) Write a Multithreading program in java to display the numberâ€™s between 1 to 100 continuously in a TextField by clicking on button. (Use Runnable Interface).

```
import java.awt.*;

import javax.swing.*;

import java.awt.event.*;

public class S17Q2 extends JFrame {
JTextField textField;

JButton startBtn;

public S17Q2() {

setTitle("Number Display");

setSize(450,300);

setDefaultCloseOperation(EXIT_ON_CLOSE);

setLayout(new FlowLayout());

textField = new JTextField(10);

add(textField);

startBtn = new JButton("Start");

startBtn.addActionListener(new ActionListener() {

public void actionPerformed(ActionEvent e) {

startDisplay();

}

});
```

```
add(startBtn);
setVisible(true);
}
private void startDisplay() {
Thread displayThread = new Thread(new DisplayRunnable());
displayThread.start();
}
private class DisplayRunnable implements Runnable {
public void run() {
for(int i=1; i<=100; i++) {
textField.setText(Integer.toString(i));
try {
Thread.sleep(100);
} catch (InterruptedException e) {
System.out.println(e);
}
}
textField.setText("");
}
}
public static void main(String args[]) {
new S17Q2();
}
}
```

Slip No– 18

Q1 [JAVA]// Repeated from Slip - 15 Q1

<!-- Slip - 18 Q2 [JAVA] -->

Q.1)Write a SERVLET program in java to accept details of student (SeatNo, Stud_Name, Class, Total_Marks). Calculate percentage and grade obtained and display details on page. -->

```
<!DOCTYPE html>

<html lang="en">

<head>

 <meta charset="UTF-8">

 <meta name="viewport" content="width=device-width, initial-scale=1.0">

 <title>Student Details Form</title>

</head>

<body>

 <h2>Enter Student Details</h2>

 <form action="S18Q2" method="post">

 <label for="seat_no">Seat No:</label>

 <input type="text" id="seat_no" name="seat_no" required><br><br>

 <label for="name">Name:</label>

 <input type="text" id="name" name="name" required><br><br>

 <label for="class">Class:</label>

 <input type="text" id="class" name="class" required><br><br>

 <label for="total_marks">Total Marks:</label>

 <input type="text" id="total_marks" name="total_marks"

required><br><br>

 <input type="submit" value="Submit">

 </form>

</body>
```

```
</html>
```

Q.2)Write a SERVLET program in java to accept details of student (SeatNo, Stud_Name, Class, Total_Marks). Calculate percentage and grade obtained and display details on page.

```
import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;

public class S18Q2 extends HttpServlet {

public void doPost(HttpServletRequest req, HttpServletResponse res) throws

IOException, ServletException {

res.setContentType("text/html");

PrintWriter out = res.getWriter();

int seatNo = Integer.parseInt(req.getParameter("seat_no"));
String name = req.getParameter("name");

String stu_class= req.getParameter("class");

int totalMarks = Integer.parseInt(req.getParameter("total_marks"));

double perc = (totalMarks / 500.0) * 100;

String grade;

if (perc >= 90) {

grade = "A+";

} else if (perc >= 80) {

grade = "A";

} else if (perc >= 70) {

grade = "B";

} else if (perc >= 60) {
```

```
grade = "C";
} else if (perc >= 50) {
grade = "D";
} else {
grade = "Fail!";
}
out.println("<html><head><title>Student
Details</title></head><body>");
out.println("<h2>Student Details</h2>");
out.println("<p>Seat No.: " + seatNo + "</p>");
out.println("<p>Name: " + name + "</p>");
out.println("<p>Class: " + stu_class + "</p>");
out.println("<p>Total Marks: " + totalMarks + "</p>");
out.println("<p>Percentage: " + perc + "</p>");
out.println("<p>Grade: " + grade + "</p>");
out.println("</body></html>");
out.close();
}
}
```

Slip No- 19

Q.1)Write a java program to accept â€˜Nâ€™ Integers from a user store them into LinkedList Collection and display only negative integers.

```
import java.util.*;
public class S19Q1 {
public static void main(String args[]) {
```

```
Scanner sc = new Scanner(System.in);

System.out.print("\nEnter the number of integers: ");

int n = sc.nextInt();
LinkedList<Integer> list = new LinkedList<>();

for(int i=0; i<n; i++) {

System.out.print("Enter integer " + (i+1) + ": ");

list.add(sc.nextInt());

}

System.out.println("\nNegative integers from list:-");

for(int num : list) {

if(num < 0) {

System.out.println(num);

}

}

sc.close();

}

}
```

Q.2)Write a SERVLET application to accept username and password, search them into database, if found then display appropriate message on the browser otherwise display error message. -->

```
<!DOCTYPE html>

<html>

 <head>

 <title>Validate User</title>

 </head>
```

```
<body>

<form action="UserValidation" method="post">

<label for="username">Enter username:</label>

<input type="text" id="username" name="username" required>

<label for="password">Enter password:</label>

<input type="password" id="password" name="password" required>

<input type="submit" value="Submit">

</form>

</body>

</html>
```

Slip No-20

1. Create a JSP page to accept a number from a user and display it in words: Example: 123 – One Two Three. The output should be in red color.

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<html>
<head>
<title>Number to Words</title>
</head>
<body>
<%
   String numStr = request.getParameter("num");
   if(numStr != null){
      char[] digits = numStr.toCharArray();
      String[] words =
{"Zero","One","Two","Three","Four","Five","Six","Seven","Eight","Nine"};
%>
      <h2 style="color:red;">
      <%
         for(char c : digits){
            if(Character.isDigit(c)){
               out.print(words[c - '0'] + " ");
            }
         }
      %>
      </h2>
<%
   }
%>
<form method="post">
```

```
Enter Number: <input type="text" name="num"/>
<input type="submit" value="Show in Words"/>
</form>
</body>
</html>
```

2. Write a java program to blink image on the JFrame continuously.

```
import javax.swing.*;
import java.awt.*;

public class BlinkImage extends JFrame {
   JLabel label;
   ImageIcon icon;

   public BlinkImage() {
      icon = new ImageIcon("yourimage.png");
      label = new JLabel(icon);
      add(label, BorderLayout.CENTER);
      setSize(400,400);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      setVisible(true);

      new Timer(500, e -> label.setVisible(!label.isVisible())).start();
   }

   public static void main(String[] args) {
      new BlinkImage();
   }
}
```

Slip No– 21

Q.1)Write a java program to accept â€˜Nâ€™ Subject Names from a user store them into LinkedList Collection and Display them by using Iterator interface.

```
import java.util.*;
public class S21Q1 {
public static void main(String args[]) {

Scanner sc = new Scanner(System.in);

LinkedList<String> list = new LinkedList<>();

System.out.print("\nEnter the number of subjects: ");

int n = sc.nextInt();
```

```
sc.nextLine();
for(int i=0; i<n; i++) {
System.out.print("Enter subject " + (i+1) + ": ");
list.add(sc.nextLine());
}
System.out.println("\nList items:-");
Iterator<String> itr = list.iterator();
while(itr.hasNext()) {
System.out.println(itr.next());
}
sc.close();
}
}
```

Q.2) Write a java program to solve producer consumer problem in which a producer produces a value and consumer consume the value before producer generate the next value. (Hint: use thread synchronization)

```
class Buffer {
private String data;
private boolean produced;
Buffer() {
this.data = null;
this.produced = false;
}
public synchronized void produce(String item) throws InterruptedException
{
```

```
// Wait until the previous value is consumed

while (produced)

wait();

// Produce the new value

data = item;

produced = true;

System.out.println("Produced: " + item);
// Notify the consumer

notify();

}

public synchronized String consume() throws InterruptedException {

// Wait until a new value is produced

while (!produced)

wait();

// Consume the produced value

String consumedItem = data;

produced = false;

System.out.println("Consumed: " + consumedItem);

// Notify the producer

notify();

return consumedItem;

}

}

class Producer extends Thread {

private String msg;

private Buffer buffer;
```

```
private int count;

Producer(String msg, Buffer buffer, int count) {

this.msg = msg;

this.buffer = buffer;

this.count = count;

}

public void run() {

try {

for (int i = 0; i < count; i++) {

buffer.produce(msg);

Thread.sleep(1000);

}

} catch (InterruptedException e) {

System.out.println(e);

}

}

}

class Consumer extends Thread {

private Buffer buffer;

private int count;
Consumer(Buffer buffer, int count) {

this.buffer = buffer;

this.count = count;

}

public void run() {

try {
```

```
for (int i = 0; i < count; i++) {

buffer.consume();

Thread.sleep(1000);

}

} catch (InterruptedException e) {

System.out.println(e);

}

}

}

public class S21Q2 {

public static void main(String args[]) {

Buffer buffer = new Buffer();

Producer producer = new Producer("Hello!", buffer, 6);

Consumer consumer = new Consumer(buffer, 6);

producer.start();

consumer.start();

}

}
```

Slip No– 22

Q.1)Write a Menu Driven program in Java for the following: Assume Employee table with attributes (ENo, EName, Salary) is already created. 1. Insert 2. Update 3. Display 4. Exit.

```
import java.sql.*;

public class S22Q1 {

static final String JDBC_DRIVER = "org.postgresql.Driver";

static final String DB_URL = "jdbc:postgresql://localhost:5432/ty92";
```

```
static final String USER = "ty92";

static final String PASS = "ty92";

public static void main(String[] args) {

Connection conn = null;

Statement stmt = null;

try {

Class.forName(JDBC_DRIVER);

conn = DriverManager.getConnection(DB_URL, USER, PASS);

stmt = conn.createStatement();
boolean exit = false;

while (!exit) {

System.out.println("\nEmployee Management System");

System.out.println("1. Insert");

System.out.println("2. Update");

System.out.println("3. Display");

System.out.println("4. Exit");

System.out.print("Enter your choice: ");

int choice = Integer.parseInt(System.console().readLine());

switch (choice) {

case 1:

insertEmployee(stmt);

break;

case 2:

updateEmployee(stmt);

break;

case 3:
```

```
displayEmployees(stmt);
break;
case 4:
exit = true;
break;
default:
System.out.println("Invalid choice! Please enter a
number between 1 and 4.");
}
}
stmt.close();
conn.close();
} catch (SQLException se) {
se.printStackTrace();
} catch (Exception e) {
e.printStackTrace();
} finally {
try {
if (stmt != null) stmt.close();
} catch (SQLException se2) {
}
try {
if (conn != null) conn.close();
} catch (SQLException se) {
se.printStackTrace();
```

```
}

}
}

static void insertEmployee(Statement stmt) throws SQLException {

System.out.println("\nInsert Employee");

System.out.print("Enter Employee Number: ");

int eno = Integer.parseInt(System.console().readLine());

System.out.print("Enter Employee Name: ");

String ename = System.console().readLine();

System.out.print("Enter Designation: ");

String designation = System.console().readLine();

System.out.print("Enter Salary: ");

double salary = Double.parseDouble(System.console().readLine());

String sql = "INSERT INTO Employee (eno, ename, designation, salary)

VALUES (" + eno + ", '" + ename + "', '" + designation + "', " + salary + ")";

stmt.executeUpdate(sql);

System.out.println("Employee inserted successfully.");

}

static void updateEmployee(Statement stmt) throws SQLException {

System.out.println("\nUpdate Employee");

System.out.print("Enter Employee Number: ");

int eno = Integer.parseInt(System.console().readLine());

System.out.print("Enter new Salary: ");

double salary = Double.parseDouble(System.console().readLine());

String sql = "UPDATE Employee SET Salary=" + salary + " WHERE eno=" +

eno;
```

```
int rowsAffected = stmt.executeUpdate(sql);

if (rowsAffected > 0)

System.out.println("Employee updated successfully.");

else

System.out.println("Employee not found.");

}

static void displayEmployees(Statement stmt) throws SQLException {

System.out.println("\nEmployee List");

String sql = "SELECT * FROM Employee";

ResultSet rs = stmt.executeQuery(sql);

while (rs.next()) {

int eno = rs.getInt("eno");

String ename = rs.getString("ename");

String designation = rs.getString("designation");

double salary = rs.getDouble("salary");

System.out.println("Employee Number: " + eno + ", Employee Name: "

+ ename + ", Designation: " + designation + ", Salary: " + salary);

}
rs.close();

}

}
```

Q.2)Write a JSP program which accepts UserName in a TextBox and greets the user according to the time on server machine. -->

```
<%@ page language="java" contentType="text/html; charset=UTF-8"

pageEncoding="UTF-8"%>
```

```
<%@ page import="pkg.GreetingService" %>

<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8">

<title>Greeting</title>

</head>

<body>

<h2>Greeting</h2>

<form action="" method="post">

Enter your name: <input type="text" name="username"><br>

Enter your password: <input type="password" name="password"><br>

<input type="submit" value="Submit">

</form>

<%

request.setCharacterEncoding("UTF-8");

String username = request.getParameter("username");

String password = request.getParameter("password");

// Check if username and password are not null or empty

if(username != null && !username.isEmpty() && password != null &&

!password.isEmpty()) {

String greeting = GreetingService.getGreeting(username);

out.println("<p>" + greeting + "</p>");

}

%>
```

```
</body>

</html>
```

Slip No- 23

Q.1)Write a java program to accept a String from a user and display each vowel from a String after every 3 seconds.

```
import java.util.*;
class VowelThread extends Thread {

private String inputString;

public VowelThread(String inputString) {

this.inputString = inputString;

}

public void run() {

try {

for(int i=0; i<inputString.length(); i++) {

char ch = inputString.charAt(i);

if(isVowel(ch)) {

System.out.println(ch);

Thread.sleep(3000);

}

}

} catch (InterruptedException e) {

System.out.println(e);

}

}

private boolean isVowel(char ch) {
```

```
ch = Character.toLowerCase(ch);

return ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u';

}

}

public class S23Q1 {

public static void main(String args[]) {

Scanner sc = new Scanner(System.in);

System.out.print("\nEnter a string: ");

String inputString = sc.nextLine();

VowelThread thread = new VowelThread(inputString);

thread.start();

sc.close();

}

}
```

Q.2) Write a java program to accept â€˜Nâ€™ student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface.

```
import java.util.*;

public class S23Q2 {

public static void main(String args[]) {
List<String> studentNames = new ArrayList<>();

for(String arg : args)

studentNames.add(arg);

System.out.println("\nStudent names using Iterator:-");

Iterator<String> itr = studentNames.iterator();
```

```
while(itr.hasNext())
System.out.println(itr.next());
System.out.println("\nStudent names using ListIterator:-");
ListIterator<String> listItr = studentNames.listIterator();
while(listItr.hasNext())
System.out.println(listItr.next());
}
}
```

Slip No- 24

Q.1)Write a java program to scroll the text from left to right continuously.

```
import javax.swing.*;
public class S24Q1 extends JFrame implements Runnable {
private JLabel label;
public S24Q1() {
setTitle("Text Scrolling");
setSize(400,100);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
label = new JLabel("This is the scrolling text!");
add(label);
Thread thread = new Thread(this);
thread.start();
setVisible(true);
}
public void run() {
```

```
try {

while(true) {

Thread.sleep(100);

String text = label.getText();

text = text.substring(1) + text.charAt(0);

label.setText(text);

}

} catch (InterruptedException e) {
System.out.println(e);

}

}

public static void main(String args[]) {

new S24Q1();

}

}
```

Q.2)Write a JSP script to accept username and password from user, if they are same then display â€œLogin Successfullyâ€ message in Login.html file, otherwise display â€œLogin Failedâ€ Message in Error.html file. -->

```
<%@ page language="java" contentType="text/html; charset=UTF-8"

pageEncoding="UTF-8" %>

<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8">

<title>Login</title>
```

```
</head>

<body>

<%

String username = request.getParameter("username");

String password = request.getParameter("password");

if(username != null && password != null &&

username.equals(password)) {

%>

<h1>Login Successfully</h1>

<%

} else {

response.sendRedirect("Error.html");

}

%>

</body>

</html>
```

Slip No-25

1. Write a JSP program to accept Name and Age of Voter and check whether he is eligible for voting or not.

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<html>
<head><title>Voter Eligibility</title></head>
<body>
<form method="post">
Name: <input type="text" name="name"/><br>
Age: <input type="text" name="age"/><br>
<input type="submit" value="Check"/>
</form>
<%
   String name = request.getParameter("name");
   String ageStr = request.getParameter("age");
   if(name != null && ageStr != null){
```

```
        int age = Integer.parseInt(ageStr);
        if(age >= 18){
            out.println("<h2 style='color:green'>" + name + " is Eligible for Voting</h2>");
        } else {
            out.println("<h2 style='color:red'>" + name + " is Not Eligible for Voting</h2>");
        }
    }
%>
</body>
</html>
```

2. Write a Java Program for the following: Assume database is already created.

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class DDLFrame extends JFrame {
    JTextField query;
    JButton createBtn, alterBtn, dropBtn;

    public DDLFrame() {
        query = new JTextField(20);
        createBtn = new JButton("Create Table");
        alterBtn = new JButton("Alter Table");
        dropBtn = new JButton("Drop Table");

        JPanel panel = new JPanel();
        panel.add(new JLabel("Type Your DDL Query Here"));
        panel.add(query);
        panel.add(createBtn);
        panel.add(alterBtn);
        panel.add(dropBtn);

        add(panel, BorderLayout.CENTER);
        setSize(400,200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);

        ActionListener listener = e -> JOptionPane.showMessageDialog(this,"Executed:
"+query.getText());
        createBtn.addActionListener(listener);
        alterBtn.addActionListener(listener);
        dropBtn.addActionListener(listener);
    }

    public static void main(String[] args) {
        new DDLFrame();
    }
}
```

Slip No- 26

Q.1)Write a Java program to delete the details of given employee (ENo EName Salary). Accept employee ID through command line. (Use PreparedStatement Interface)

```
import java.sql.*;

public class S26Q1 {

public static void main(String args[]) {
int empId = Integer.parseInt(args[0]);

try {

Class.forName("org.postgresql.Driver");

Connection con =

DriverManager.getConnection("jdbc:postgresql://localhost:5432/ty92","ty92","ty

92");

PreparedStatement pstmt = con.prepareStatement("DELETE FROM

Employee WHERE eno = ?");

pstmt.setInt(1, empId);

int rowAffected = pstmt.executeUpdate();

if(rowAffected > 0)

System.out.println("\nDetails of employee with ID " + empId +

" deleted successfully!");

else

System.out.println("Employee with ID " + empId + " not

found.");

} catch(SQLException | ClassNotFoundException e) {

System.out.println(e);
```

```
}

}

}
```

Q.2) Write a JSP program to calculate sum of first and last digit of a given number. Display sum in Red Color with font size

```
<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8">

<title>Sum of first and last digit</title>

</head>

<body>

<h2>Calculate Sum of First and Last Digits</h2>

<form action="" method="post">

Enter a number: <input type="text" name="number"><br>

<input type="submit" value="Calculate">

</form>

<%

String numberStr = request.getParameter("number");

if(numberStr != null && !numberStr.isEmpty()) {

int number = Integer.parseInt(numberStr);

int firstDigit = Character.getNumericValue(numberStr.charAt(0));

int lastDigit = number % 10;

int sum = firstDigit + lastDigit;

%>

<p style="color: red; font-size: 18px;">Sum of first and last digit of <%=
```

```
number %>: <%= sum %></p>

<% } %>

</body>

</html>
```

Slip No- 27

Q.1)Write a Java Program to display the details of College (CID, CName, address, Year) on JTable.

```
import javax.swing.*;

import javax.swing.table.DefaultTableModel;

import java.awt.*;

import java.util.ArrayList;

public class S27Q1 extends JFrame {

public S27Q1() {

setTitle("College Details");

setSize(600, 400);

setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

setLocationRelativeTo(null);

initComponents();

}
private void initComponents() {

// Sample data for colleges

ArrayList<College> colleges = new ArrayList<>();

colleges.add(new College(1, "ABC College", "123 Main St", 2000));

colleges.add(new College(2, "XYZ College", "456 Elm St", 1995));

colleges.add(new College(3, "PQR College", "789 Oak St", 2010));
```

```
// Create JTable with DefaultTableModel
String[] columnNames = {"CID", "CName", "Address", "Year"};
DefaultTableModel model = new DefaultTableModel(columnNames, 0);
for (College college : colleges) {
Object[] rowData = {college.getCID(), college.getCName(),
college.getAddress(), college.getYear()};
model.addRow(rowData);
}
JTable table = new JTable(model);
// Add JTable to JScrollPane
JScrollPane scrollPane = new JScrollPane(table);
getContentPane().add(scrollPane, BorderLayout.CENTER);
}
public static void main(String[] args) {
new S27Q1().setVisible(true);
}
}
class College {
private int CID;
private String CName;
private String address;
private int year;
public College(int CID, String CName, String address, int year) {
this.CID = CID;
this.CName = CName;
```

```
this.address = address;

this.year = year;

}

public int getCID() {

return CID;

}

public String getCName() {
return CName;

}

public String getAddress() {

return address;

}

public int getYear() {

return year;

}

}
```

Q.2) Write a SERVLET program to change inactive time interval of session.

```
import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;

public class S27Q2 extends HttpServlet {

protected void doGet(HttpServletRequest request, HttpServletResponse

response)

throws ServletException, IOException {

// Get the current session or create a new one if it doesn't exist
```

```
HttpSession session = request.getSession();
// Set the inactive time interval of the session to 5 minutes (300
seconds)
session.setMaxInactiveInterval(300);
// Set response content type
response.setContentType("text/html");
PrintWriter out = response.getWriter();
out.println("<html>");
out.println("<head>");
out.println("<title>Session Timeout Interval Changed</title>");
out.println("</head>");
out.println("<body>");
out.println("<h1>Session Timeout Interval Changed</h1>");
out.println("<p>The inactive time interval of the session has been
changed to 5 minutes.</p>");
out.println("</body>");
out.println("</html>");
}
}
```

Slip No- 28

Q.1) Write a JSP script to accept a String from a user and display it in reverse order. -->

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>
<!DOCTYPE html>
```

```
<html>
<head>
<meta charset="UTF-8">
<title>Reverse String</title>
</head>
<body>
<h2>Enter a String:</h2>
<form action="" method="post">
<input type="text" name="inputString">
<input type="submit" value="Reverse">
</form>
<%
// Get the input string from the request parameter
String inputString = request.getParameter("inputString");
// Check if the input string is not null or empty
if (inputString != null && !inputString.isEmpty()) {
// Reverse the input string
StringBuilder reversedString = new
StringBuilder(inputString).reverse();
// Display the reversed string
%>
<h2>Reversed String:</h2>
<p><%= reversedString.toString() %></p>
<%
}
```

```
%>

</body>

</html>
```

Q2 .Write a java program to display name of currently executing Thread in multithreading.

```
public class S28Q2 {

public static void main(String[] args) {

// Create and start a new thread

Thread thread = new Thread(new MyRunnable());

thread.start();
// Display the name of the main thread

System.out.println("Main thread name: " +

Thread.currentThread().getName());

}

}

class MyRunnable implements Runnable {

public void run() {

System.out.println("Currently executing thread name: " +

Thread.currentThread().getName());

}

}
```

slip No-29

Q.1) Write a Java program to display information about all columns in the DONAR

table using ResultSetMetaData.

```
import java.sql.*;
```

```
public class S29Q1 {
public static void main(String args[]) {
try {
Class.forName("org.postgresql.Driver");
Connection con =
DriverManager.getConnection("jdbc:postgresql://localhost:5432/ty92","ty92","ty
92");
Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery("SELECT * FROM Donor");
ResultSetMetaData rsmd = rs.getMetaData();
for(int i = 1; i <= rsmd.getColumnCount(); i++) {
System.out.println("Column Name: " + rsmd.getColumnName(i));
System.out.println("Data Type: " + rsmd.getColumnTypeName(i));
System.out.println("Column Type: " + rsmd.getColumnType(i));
System.out.println("----------------------------------------
--------");
}
} catch (SQLException | ClassNotFoundException e) {
System.out.println(e);
}
}
}
```

Q.2)Write a Java program to create LinkedList of integer objects and perform the following:

```
// i. Add element at first position
```

```
// ii. Delete last element

// iii. Display the size of link list

import java.util.*;

public class S29Q2 {

public static void main(String args[]) {

LinkedList<Integer> list = new LinkedList<>();

list.add(20);

list.add(30);

System.out.println("\nOriginal Liked List: " + list);

list.addFirst(10);

System.out.println("\nLiked List after adding element at first

position: " + list);

list.removeLast();

System.out.println("\nLiked List after deleting the last element: " +

list);

System.out.println("\nSize of the Linked List: " + list.size());

}

}
```

Slip No- 30

Q.1) Write a java program for the implementation of synchronization.

```
class SharedResource {

private int value = 0;

public synchronized void increment() {

value++;
```

```
System.out.println("Value incremented to: " + value);

}

public synchronized void decrement() {

value--;

System.out.println("Value decremented to: " + value);

}

}

class IncrementThread extends Thread {

private SharedResource res;

public IncrementThread(SharedResource res) {

this.res = res;

}
public void run() {

for(int i=0; i<5; i++) {

res.increment();

}

}

}

class DecrementThread extends Thread {

private SharedResource res;

public DecrementThread(SharedResource res) {

this.res = res;

}

public void run() {

for(int i=0; i<5; i++) {

res.decrement();
```

```
}

}

}

public class S30Q1 {

public static void main(String args[]) {

SharedResource res = new SharedResource();

IncrementThread Thread1 = new IncrementThread(res);

DecrementThread Thread2 = new DecrementThread(res);

Thread1.start();

Thread2.start();

}

}
```

2. Write a Java Program for the implementation of scrollable ResultSet. Assume Teacher table with attributes (TID, TName, Salary) is already created.

```
import java.sql.*;

import java.awt.*;

import javax.swing.*;

import javax.swing.table.DefaultTableModel;

public class S30Q2 extends JFrame {

private JTable table;

public S30Q2() {

super("Teacher Information");

// Create a panel for holding the table

JPanel panel = new JPanel(new BorderLayout());

getContentPane().add(panel);

// Create a scroll pane for the table

JScrollPane scrollPane = new JScrollPane();
```

```
panel.add(scrollPane, BorderLayout.CENTER);

// Create a table model

DefaultTableModel model = new DefaultTableModel();

// Set the column names

model.setColumnIdentifiers(new String[]{"Teacher ID", "Name",

"Subject"});

// Create the table with the model

table = new JTable(model);

// Set table properties

table.setAutoResizeMode(JTable.AUTO_RESIZE_ALL_COLUMNS);

table.setFillsViewportHeight(true);

// Add the table to the scroll pane

scrollPane.setViewportView(table);

// Populate the table

try {

Class.forName("org.postgresql.Driver");

Connection con =

DriverManager.getConnection("jdbc:postgresql://localhost:5432/ty92","ty92","ty

92");

Statement stmt =

con.createStatement(ResultSet.TYPE_SCROLL_INSENSITIVE,

ResultSet.CONCUR_READ_ONLY);

ResultSet rs = stmt.executeQuery("SELECT * FROM Teacher");

// Populate the table model with data from the ResultSet

while (rs.next()) {
```

```
int no = rs.getInt("tno");

String name = rs.getString("tname");

String subject = rs.getString("subject");

model.addRow(new Object[]{no, name, subject});

}

} catch (SQLException | ClassNotFoundException e) {

System.out.println(e);

}

// Set frame properties

setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

setSize(600, 400);

setLocationRelativeTo(null);

setVisible(true);

}

public static void main(String[] args) {

new S30Q2();

}

}
```